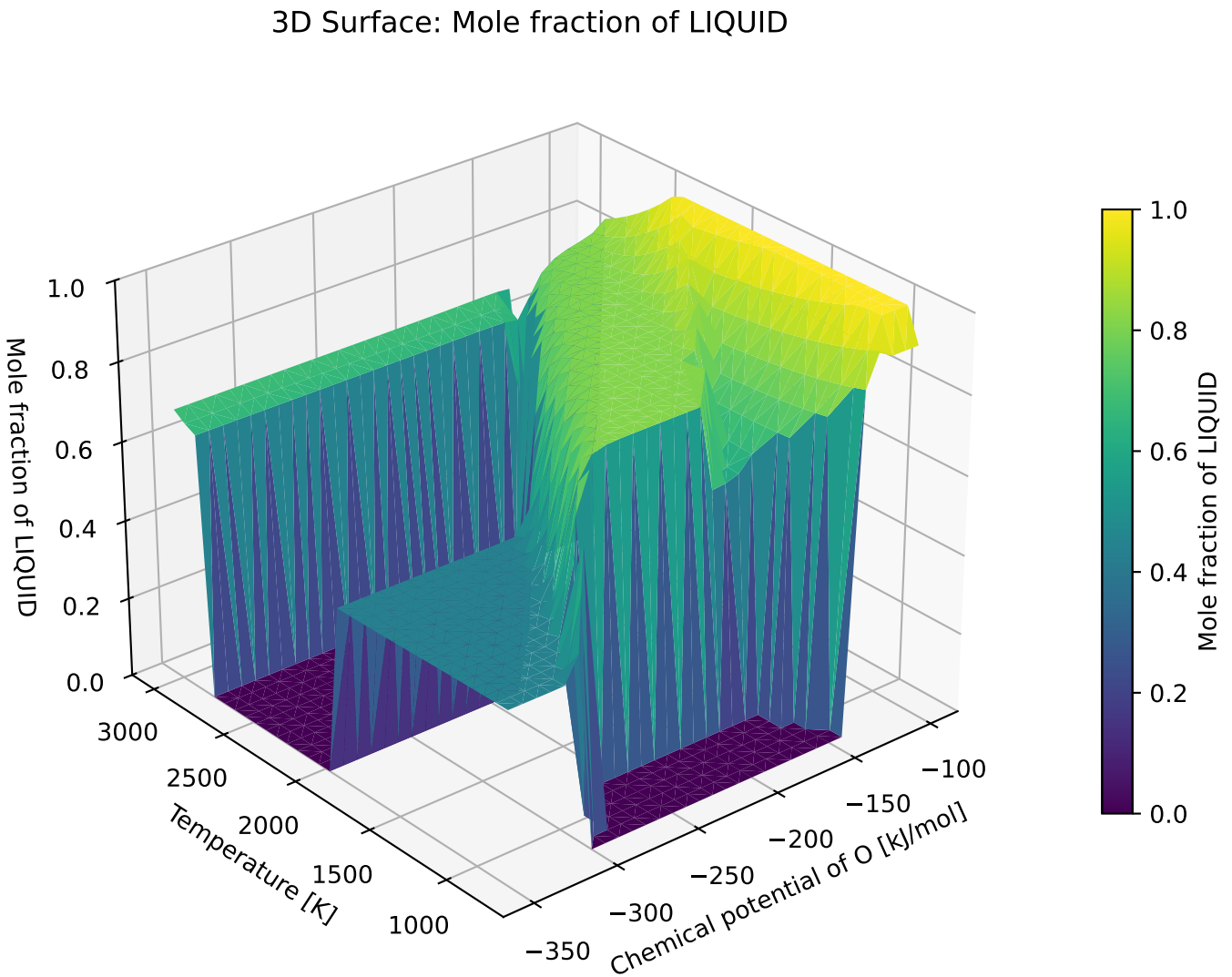
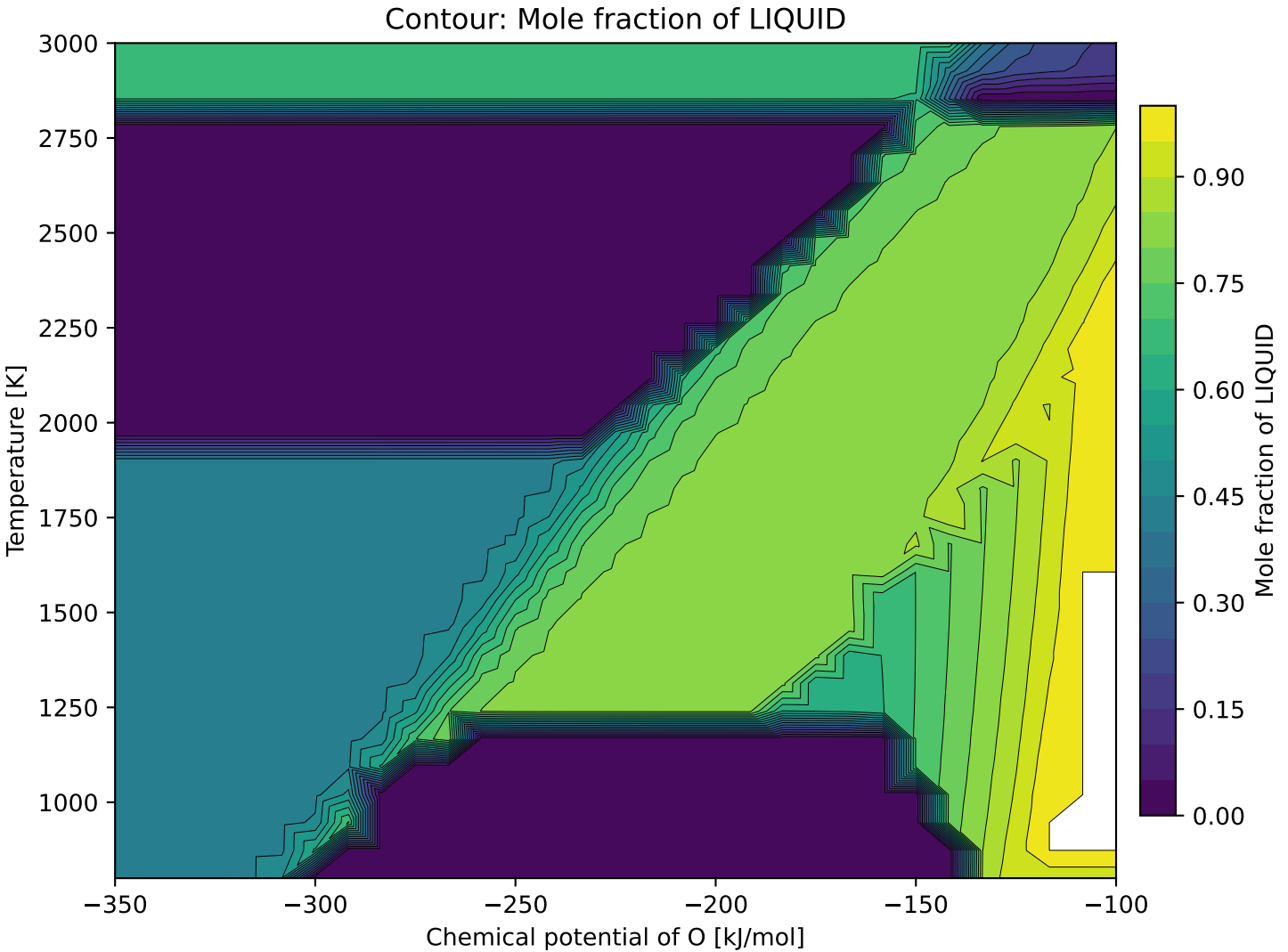
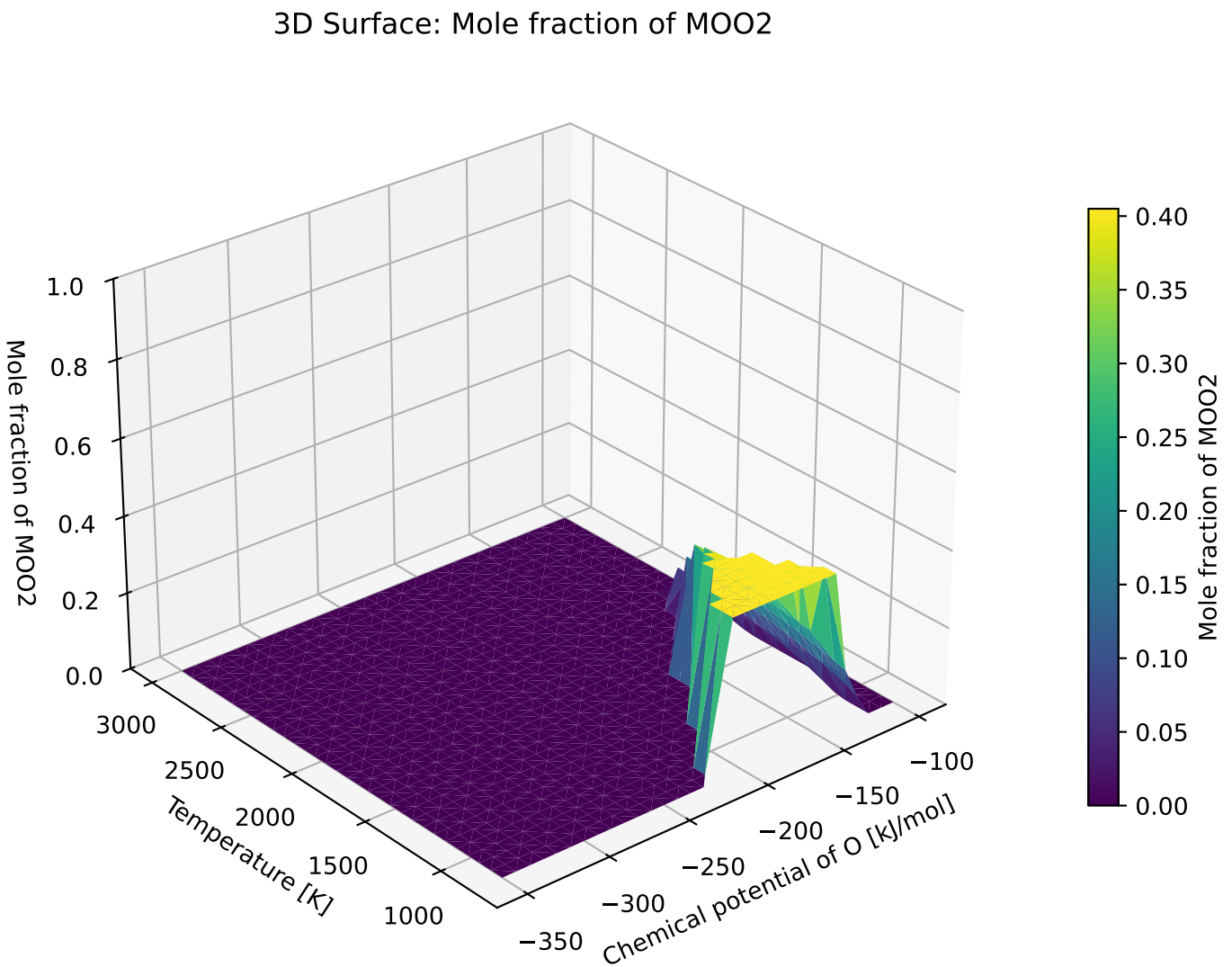
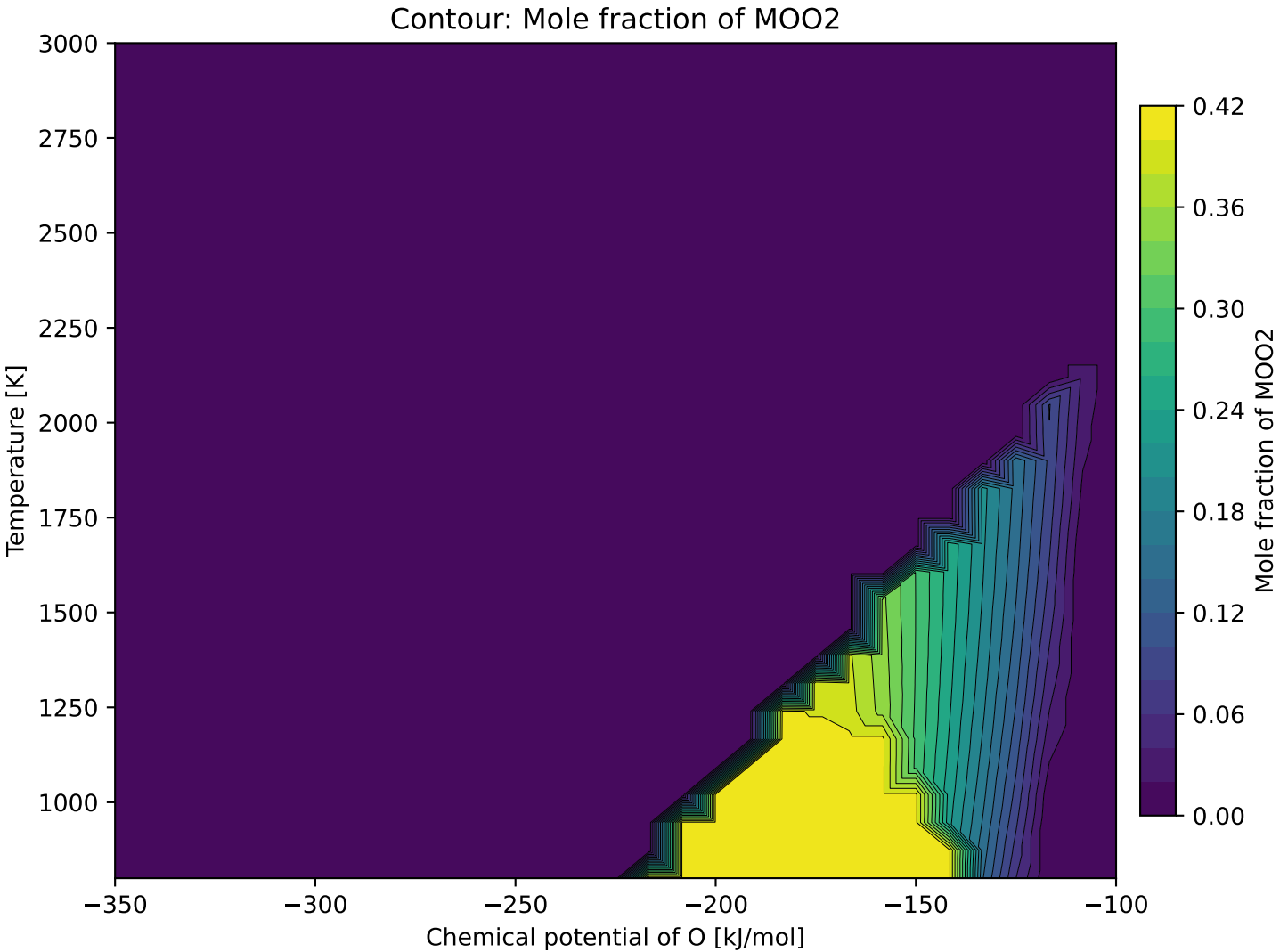
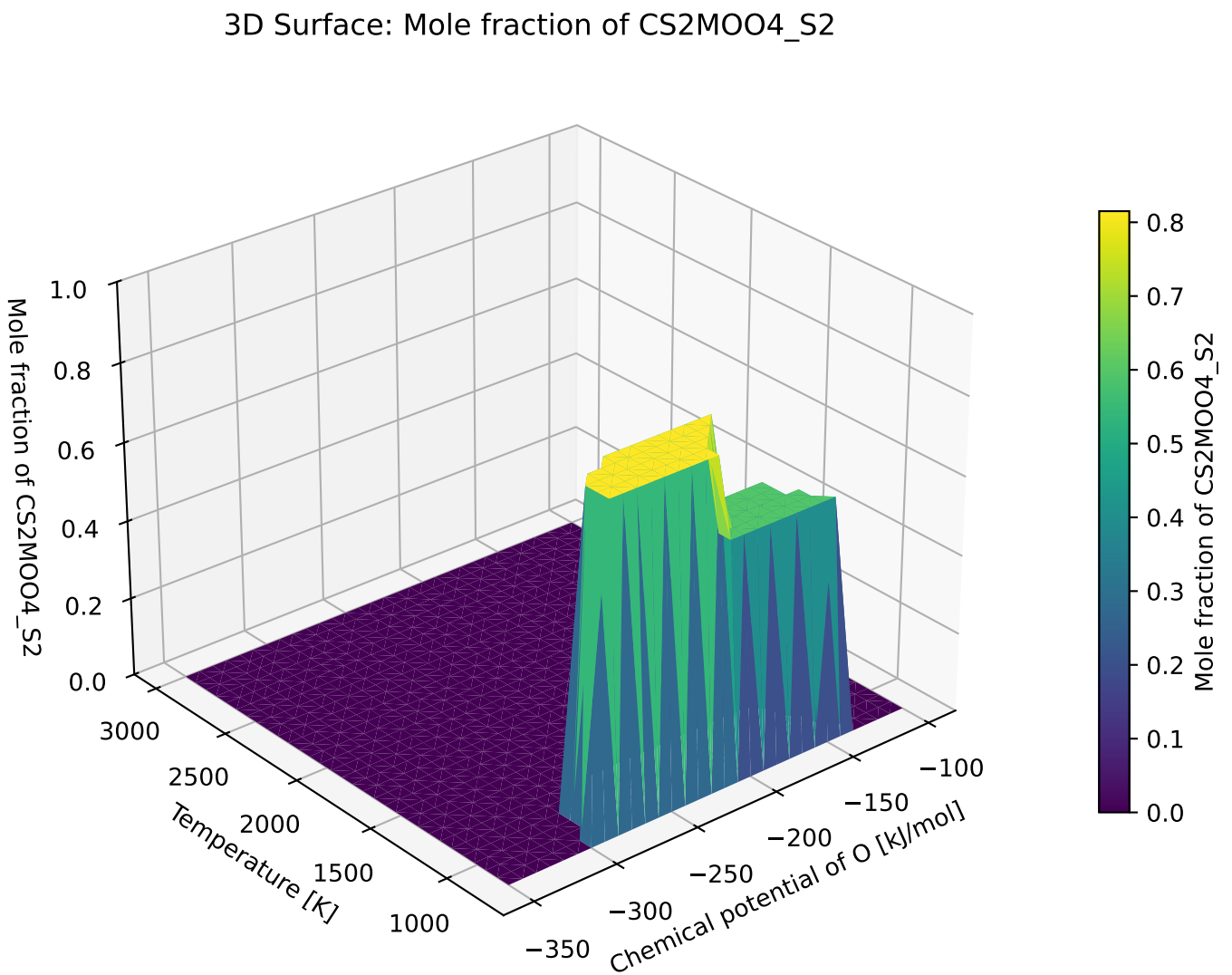
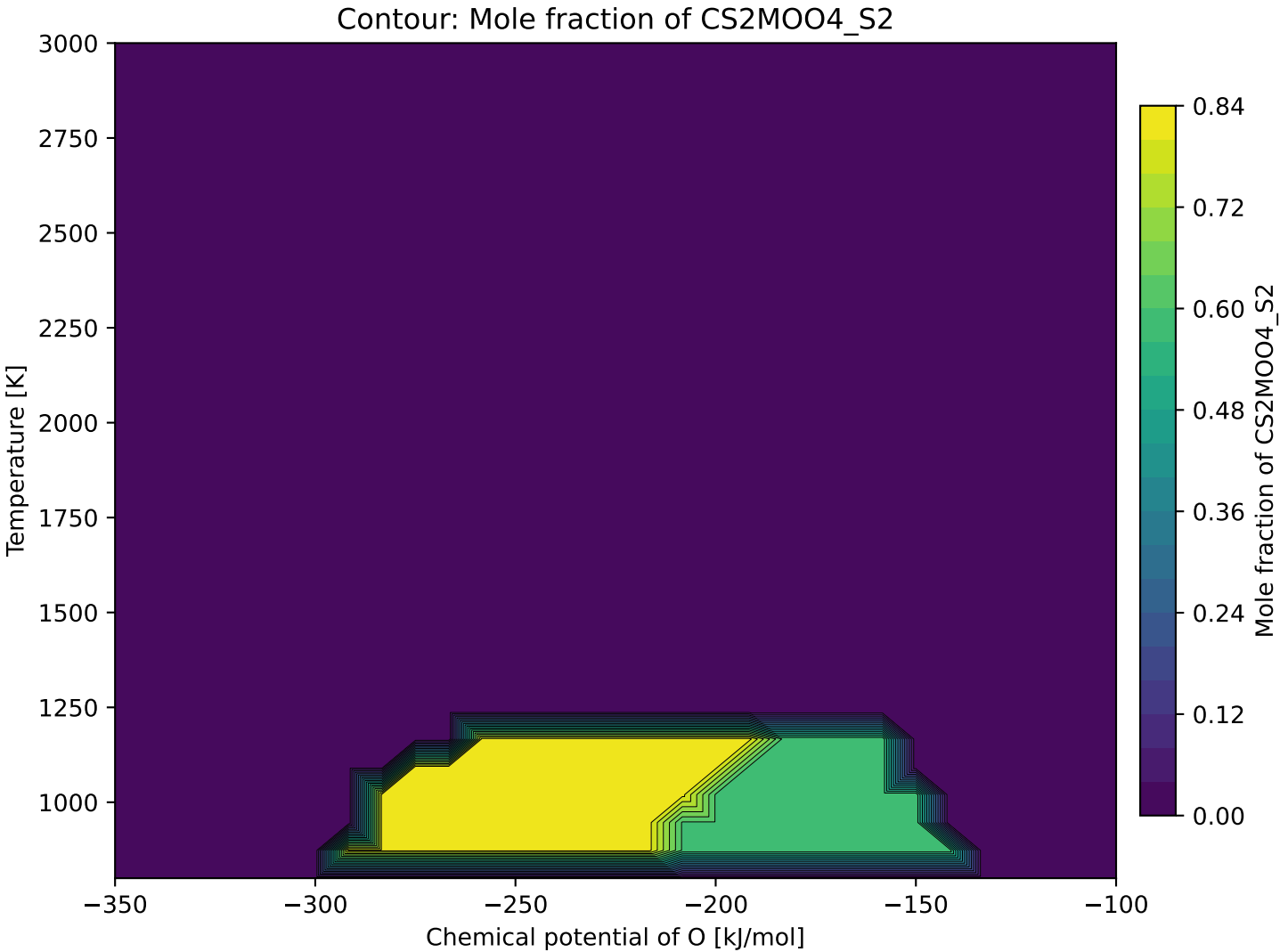

3D Surface: Mole fraction of LIQUID
Mole fraction of LIQUID
1.0
0.8
0.6
0.4
0.2
0.0
Temperature [K]
3000
2500
2000
1500
1000
Chemical potential of O [kJ/mol]
−350
−300
−250
−200
−150
−100
Mole fraction of LIQUID
1.0
0.8
0.6
0.4
0.2
0.0

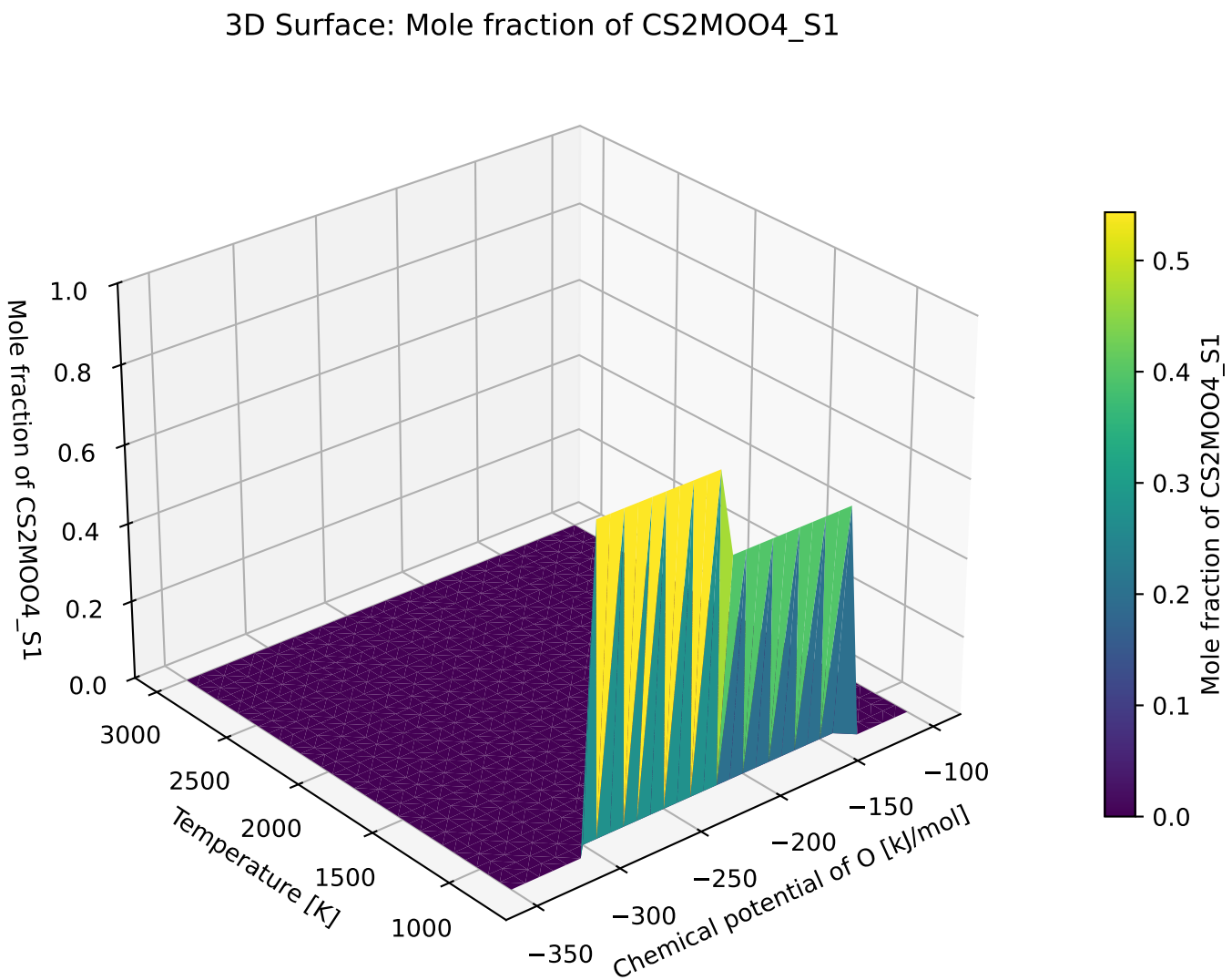

Contour: Mole fraction of LIQUID
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of LIQUID
0.90
0.75
0.60
0.45
0.30
0.15
0.00

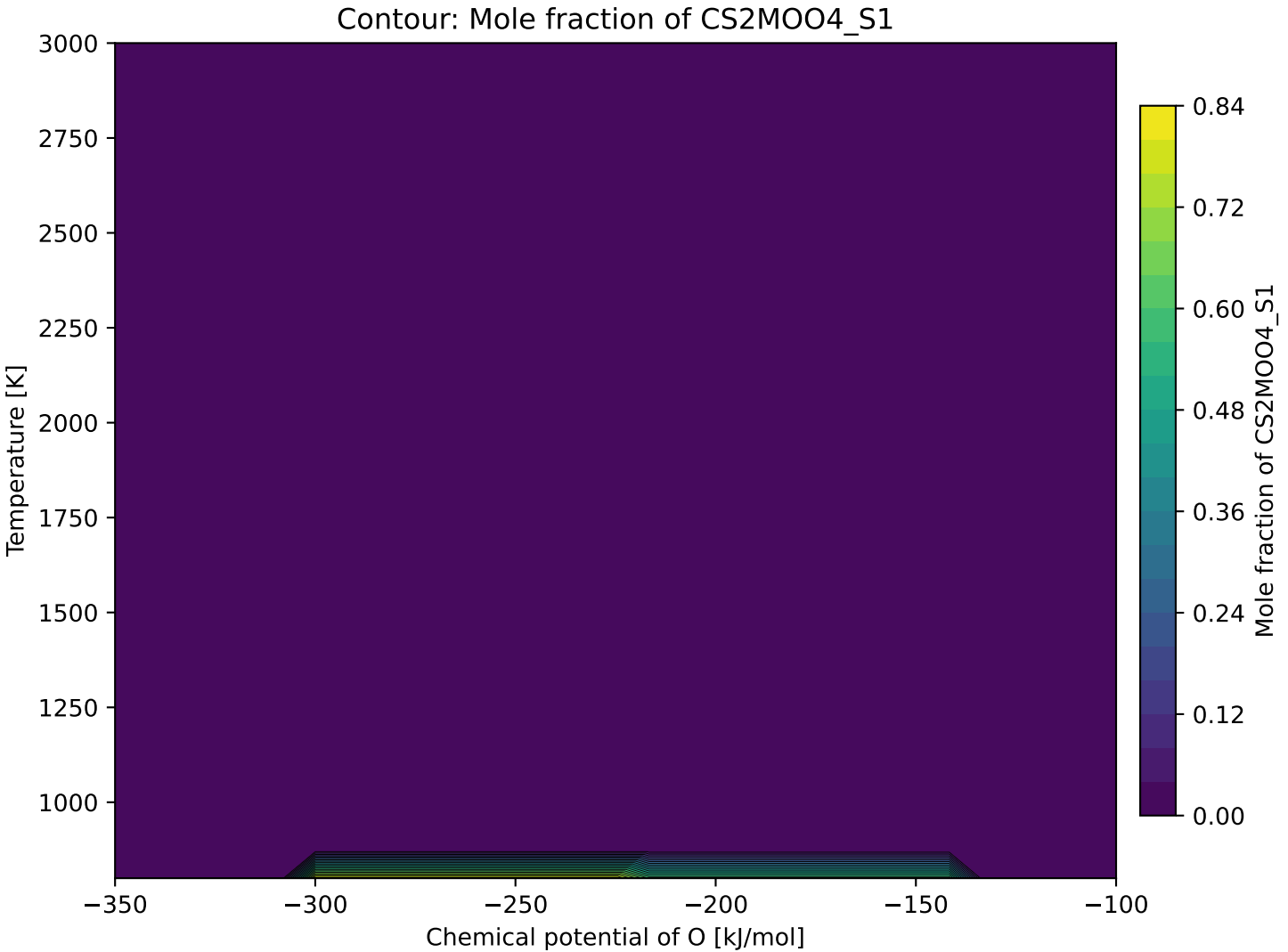

3D Surface: Mole fraction of MOO2
Mole fraction of MOO2
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Mole fraction of MOO2

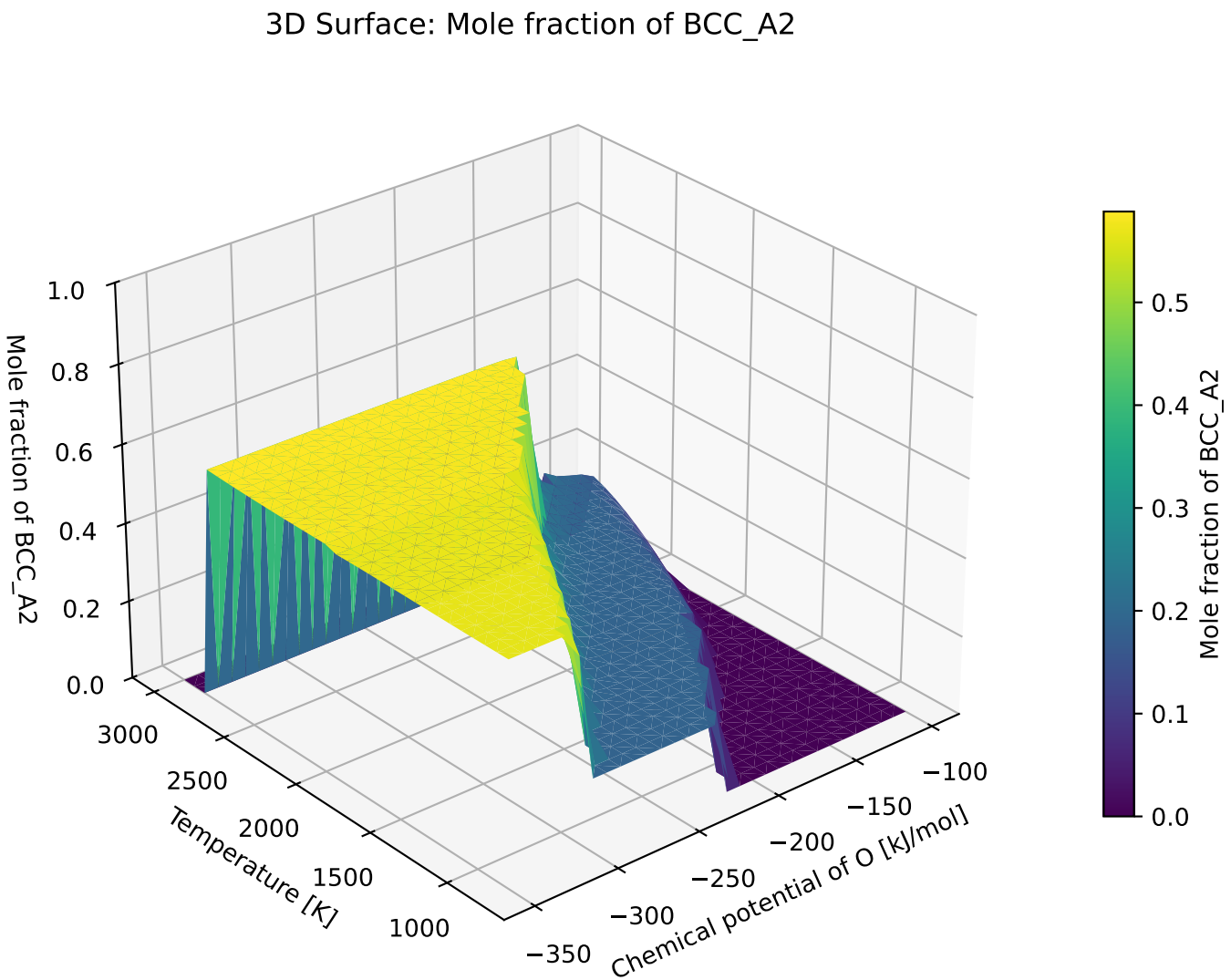

Contour: Mole fraction of MOO2
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of MOO2
0.42
0.36
0.30
0.24
0.18
0.12
0.06
0.00

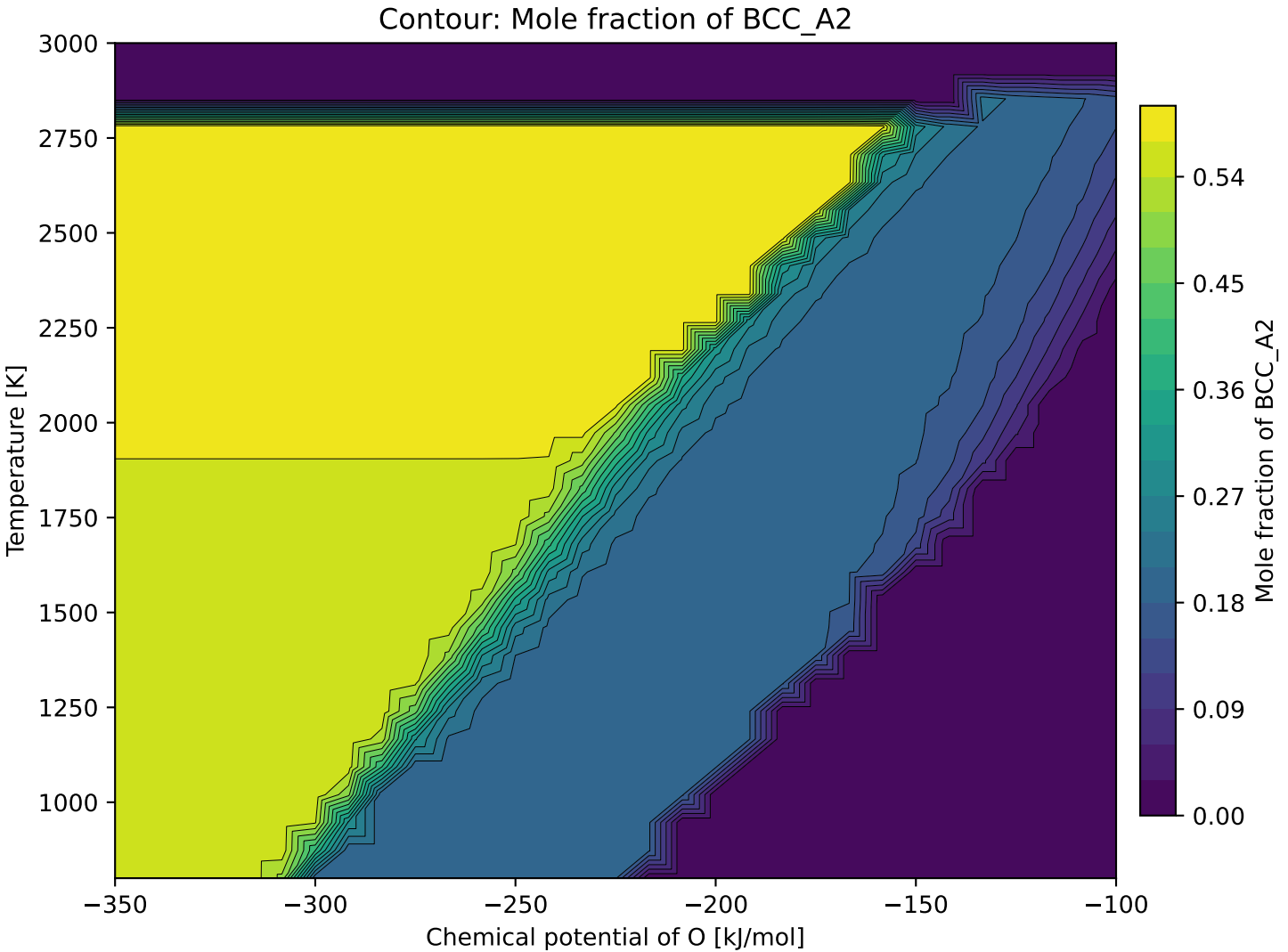

3D Surface: Mole fraction of CS2MOO4_S2
Mole fraction of CS2MOO4_S2
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Mole fraction of CS2MOO4_S2

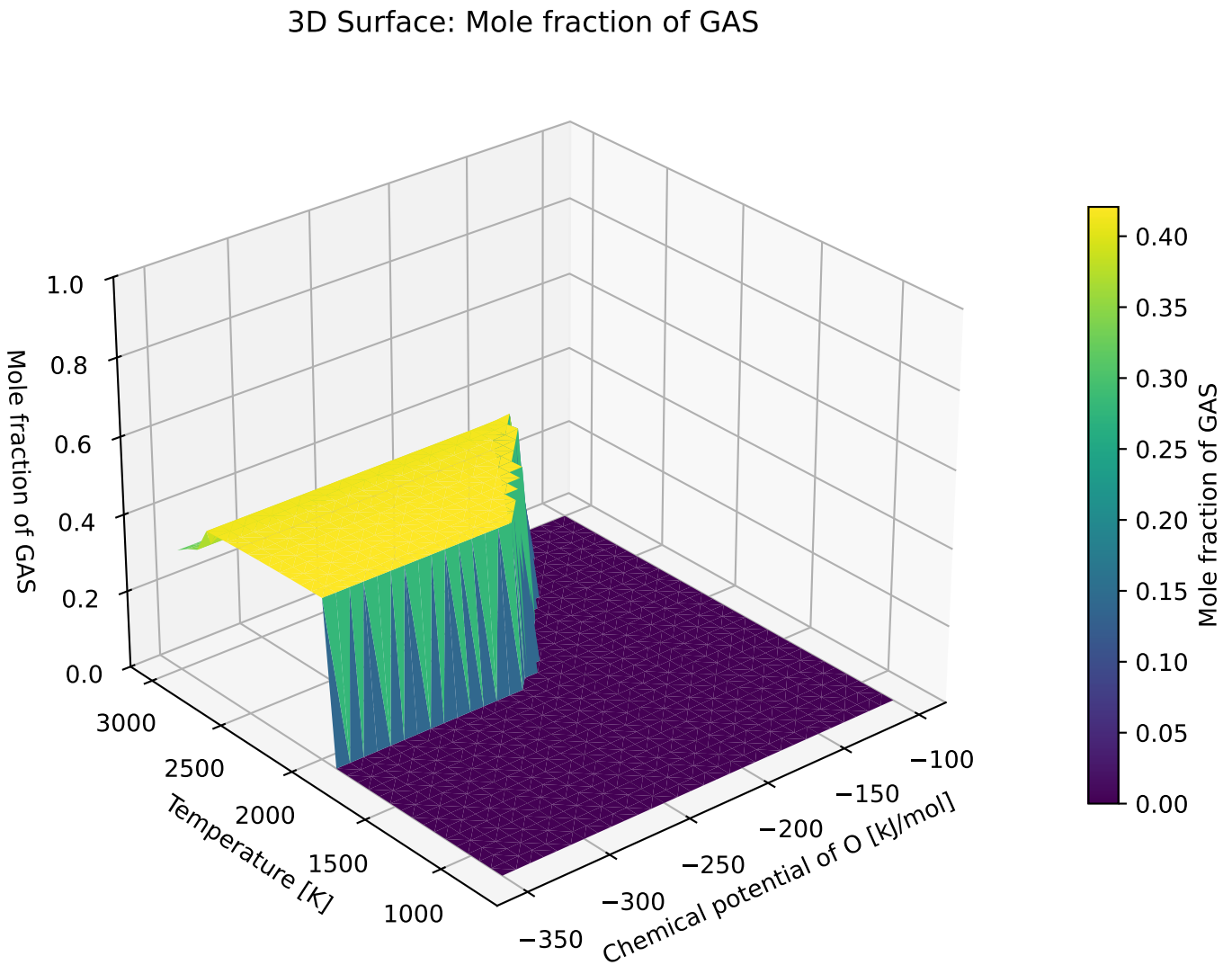

Contour: Mole fraction of CS2MOO4_S2
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of CS2MOO4_S2
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

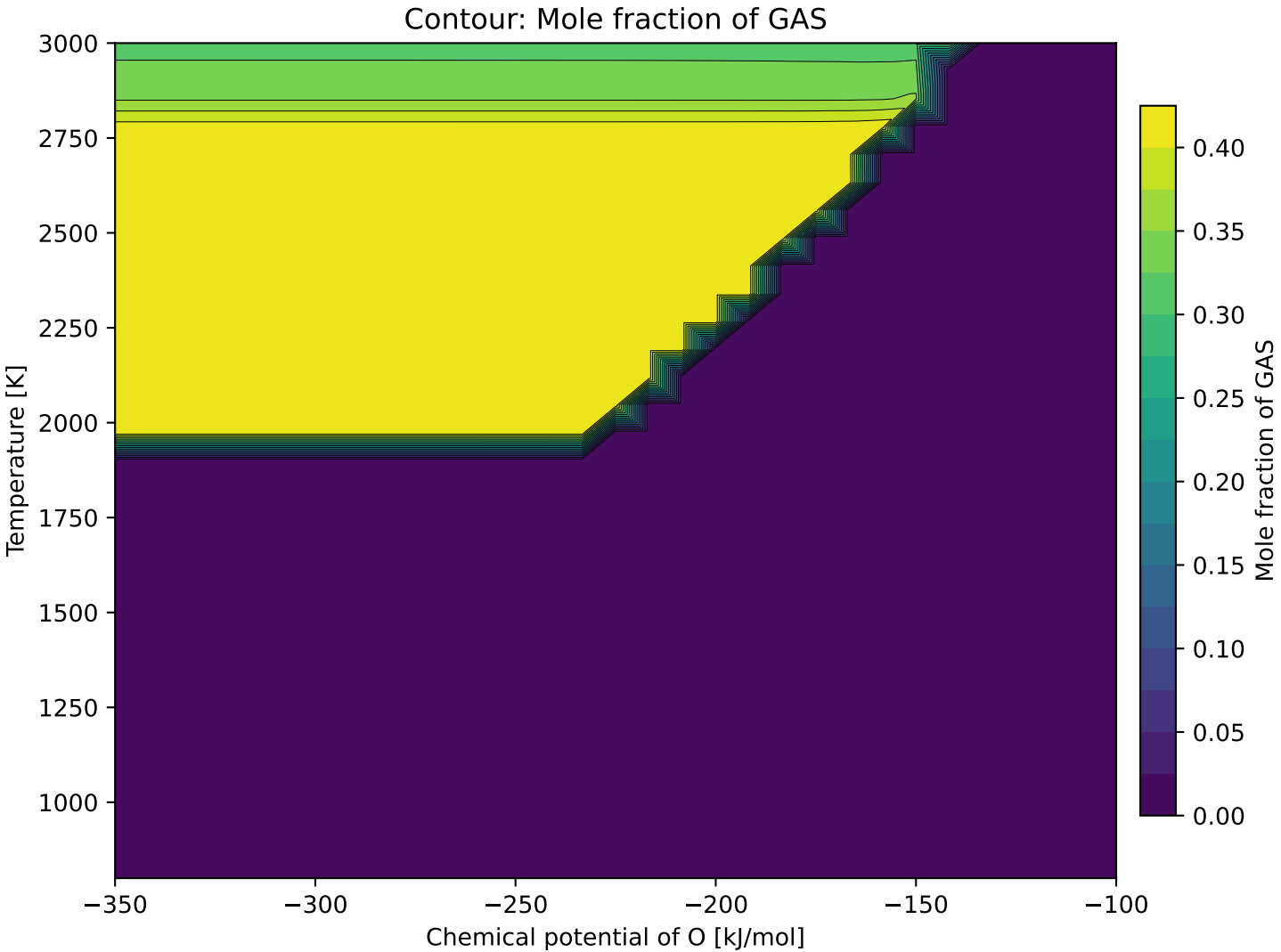

3D Surface: Mole fraction of CS2MOO4_S1
Mole fraction of CS2MOO4_S1
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.5
0.4
0.3
0.2
0.1
0.0
Mole fraction of CS2MOO4_S1

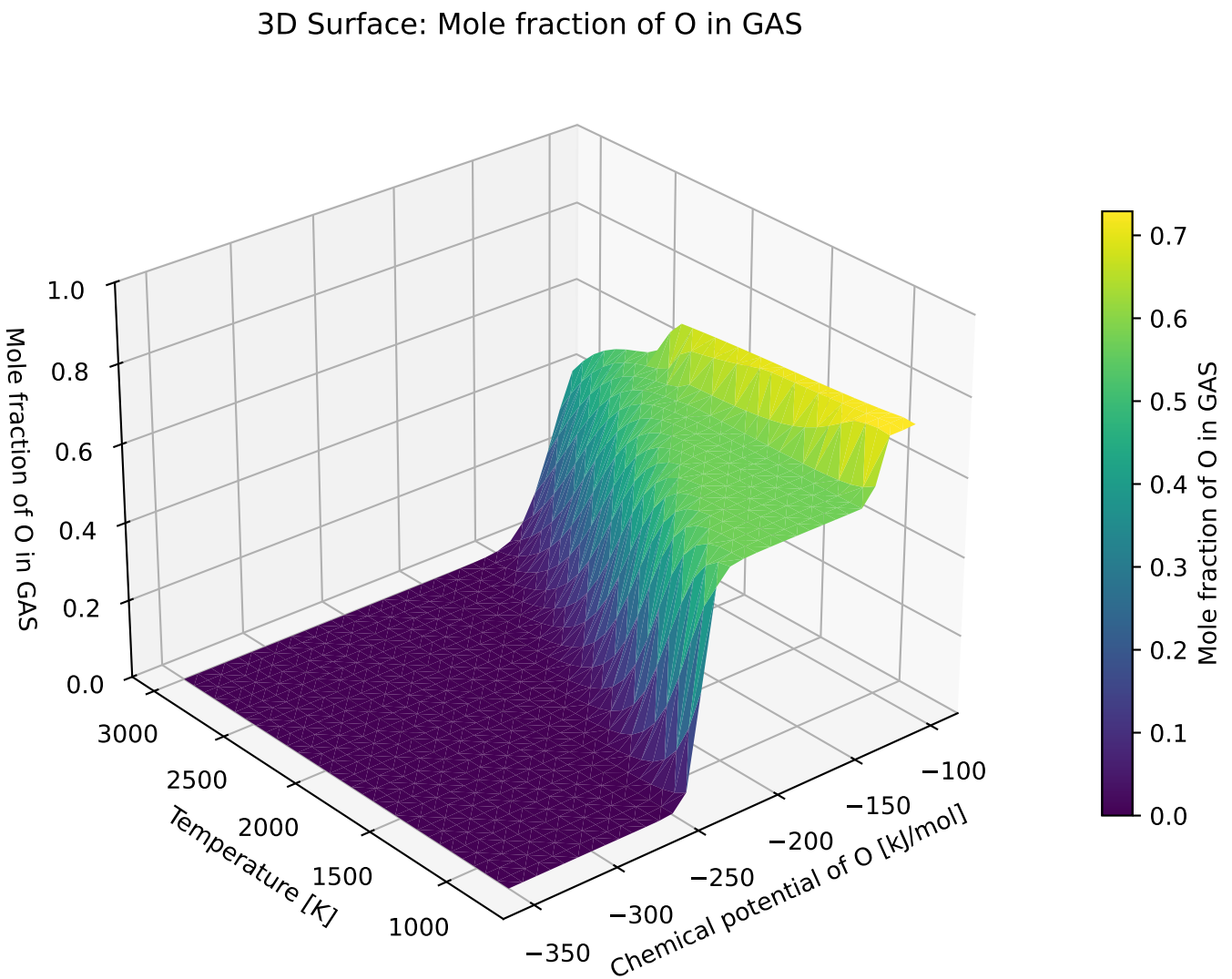

Contour: Mole fraction of CS2MOO4_S1
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of CS2MOO4_S1
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00

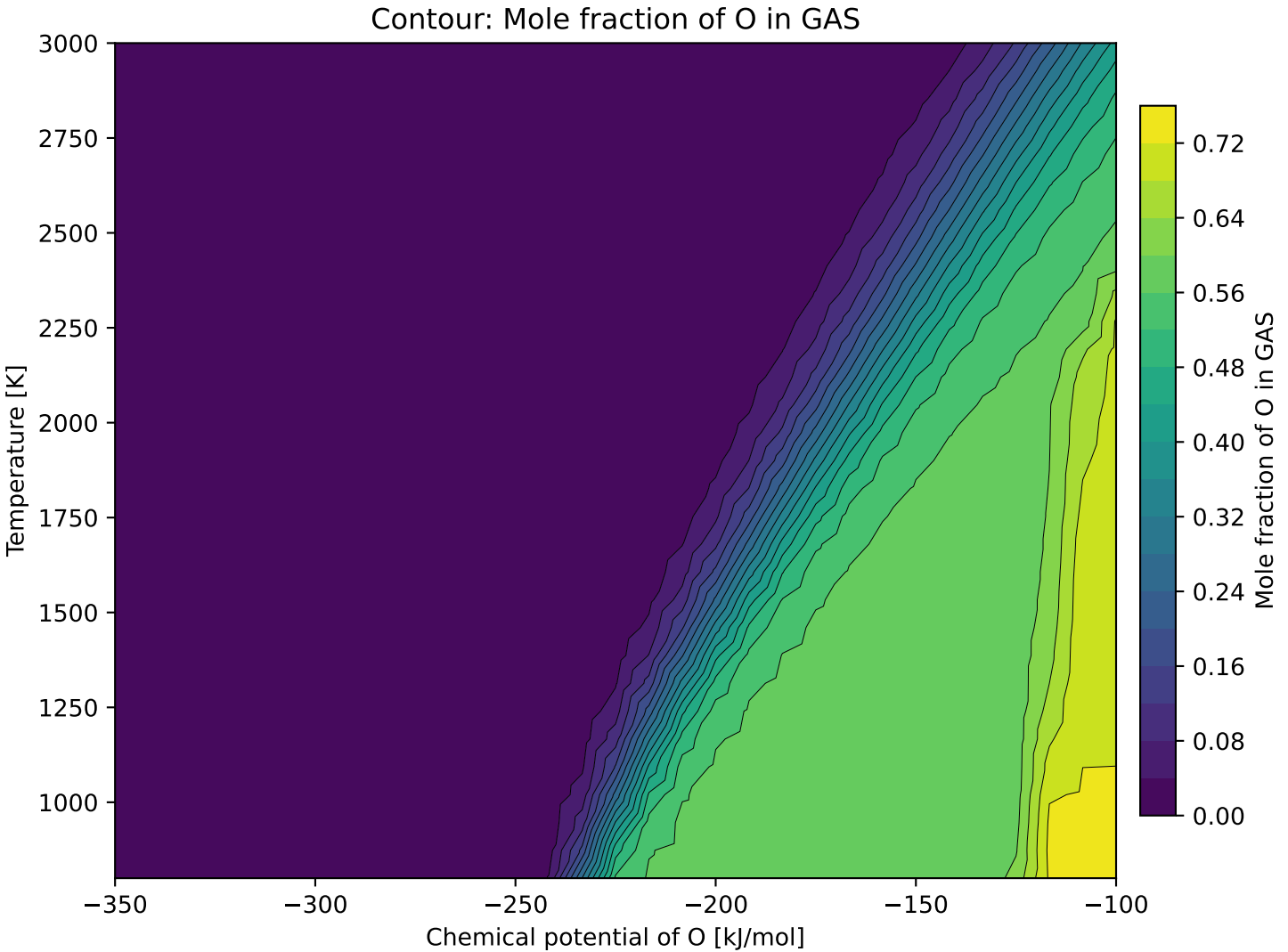

3D Surface: Mole fraction of BCC_A2
Mole fraction of BCC_A2
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.5
0.4
0.3
0.2
0.1
0.0
Mole fraction of BCC_A2

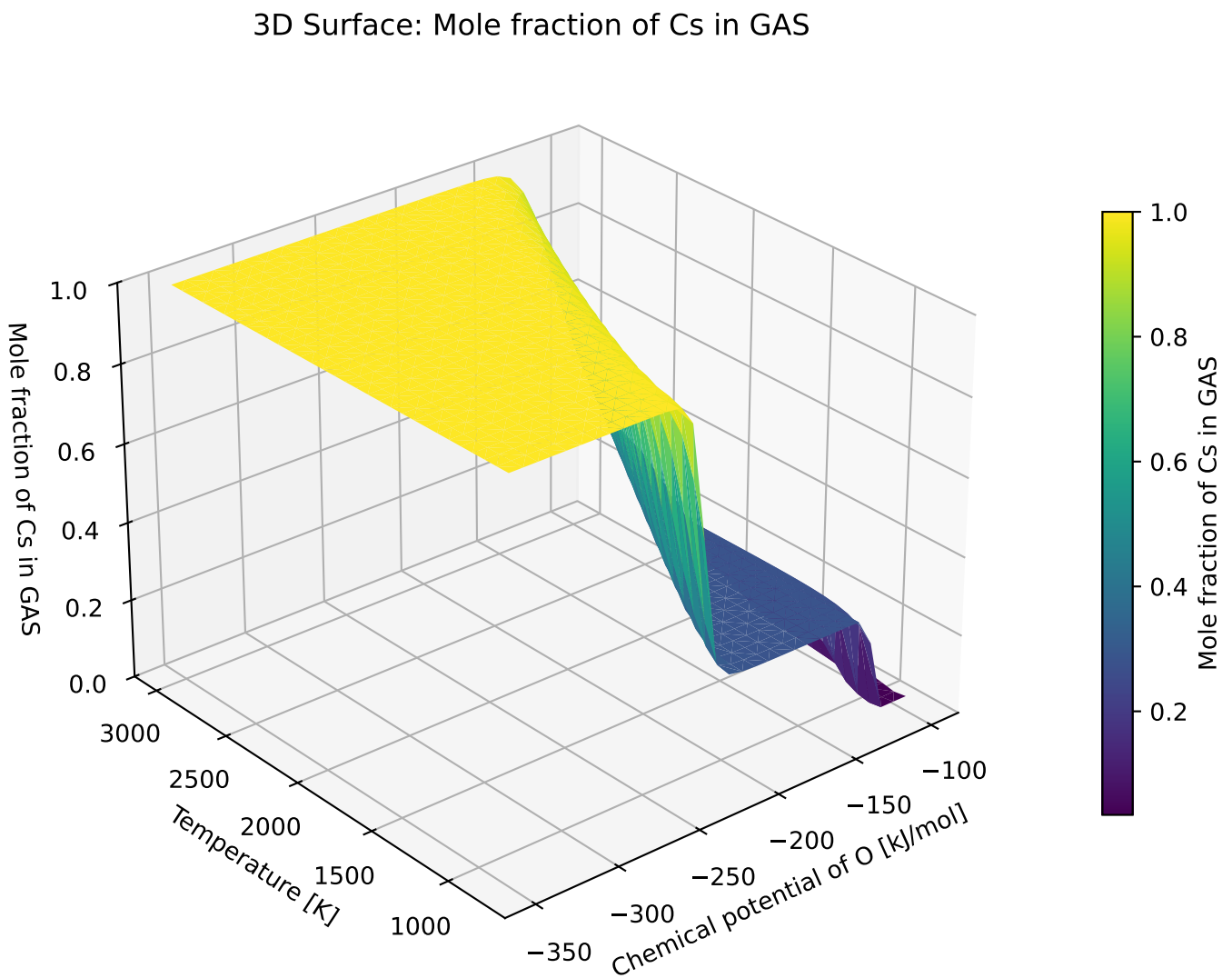

Contour: Mole fraction of BCC_A2
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of BCC_A2
0.54
0.45
0.36
0.27
0.18
0.09
0.00

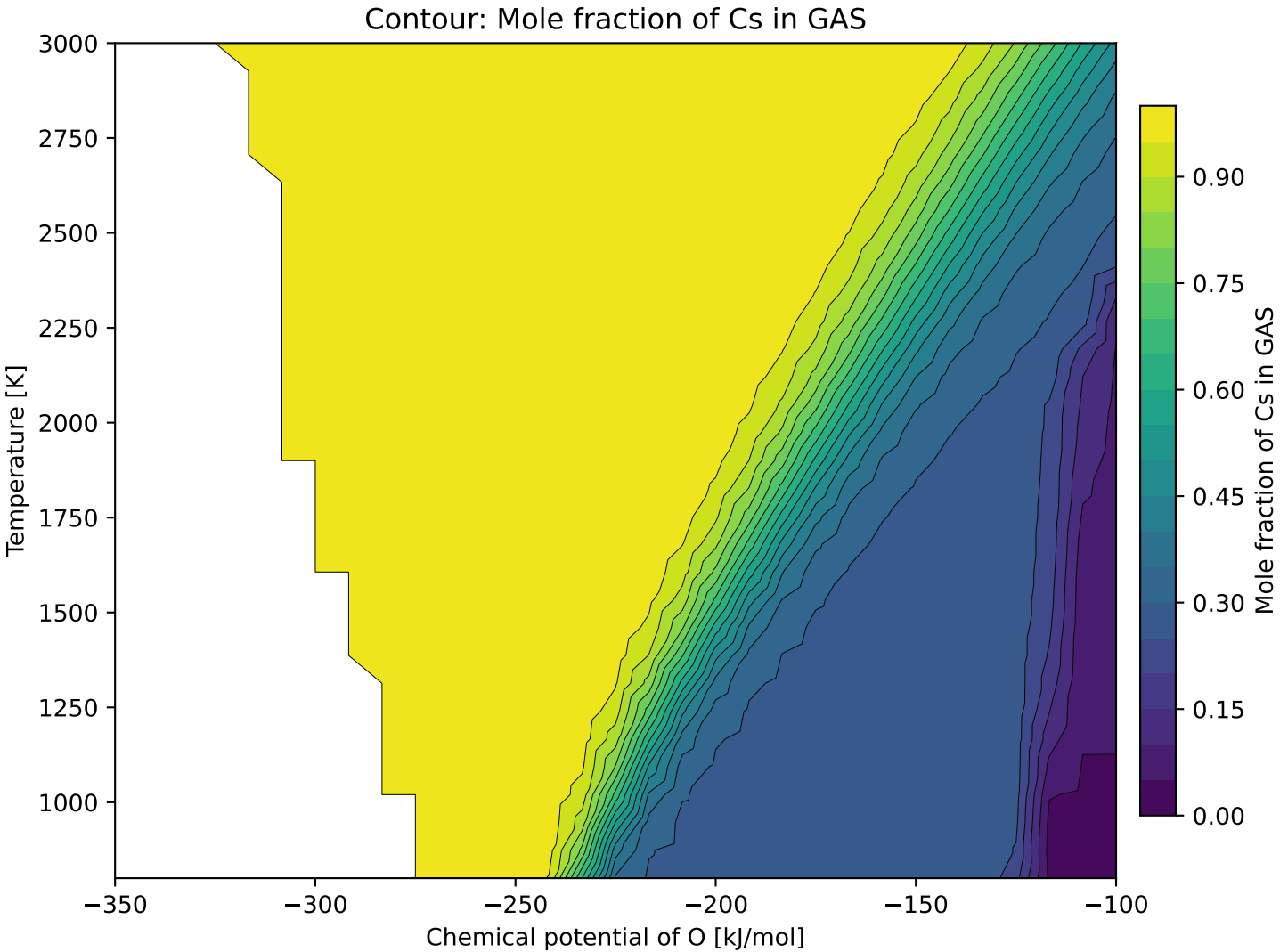

3D Surface: Mole fraction of GAS
Mole fraction of GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Mole fraction of GAS

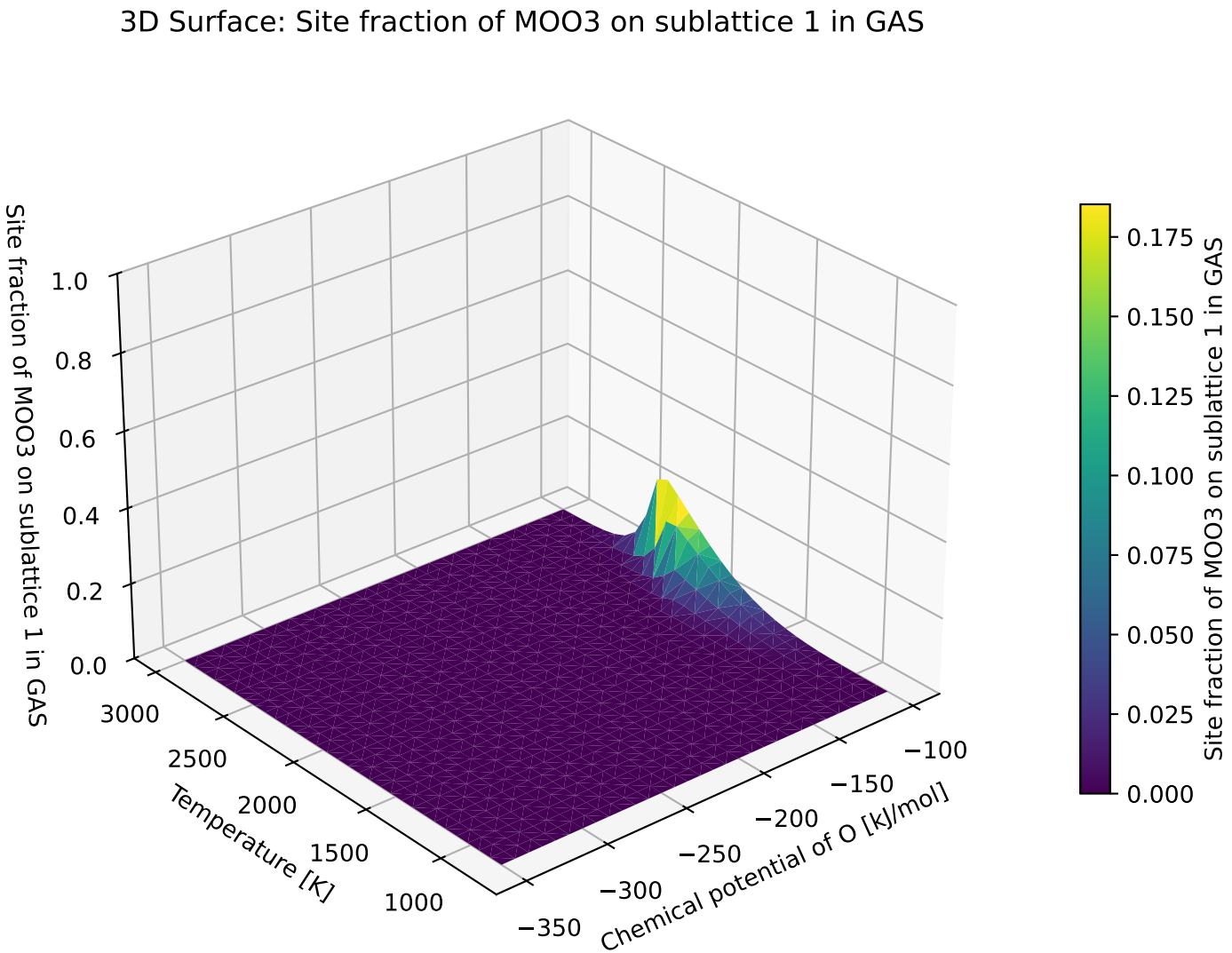

Contour: Mole fraction of GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of GAS
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00

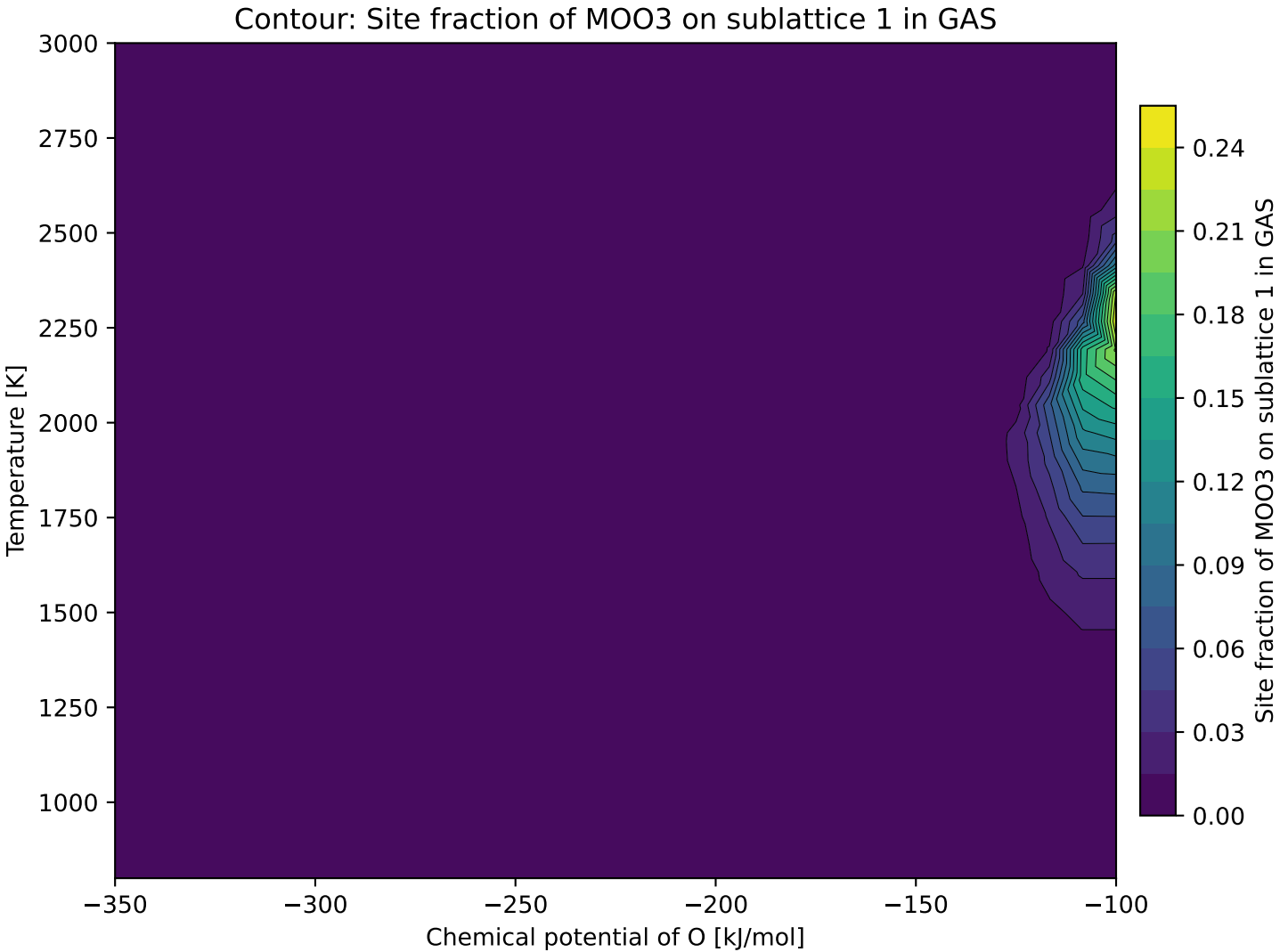

3D Surface: Mole fraction of O in GAS
Mole fraction of O in GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Mole fraction of O in GAS

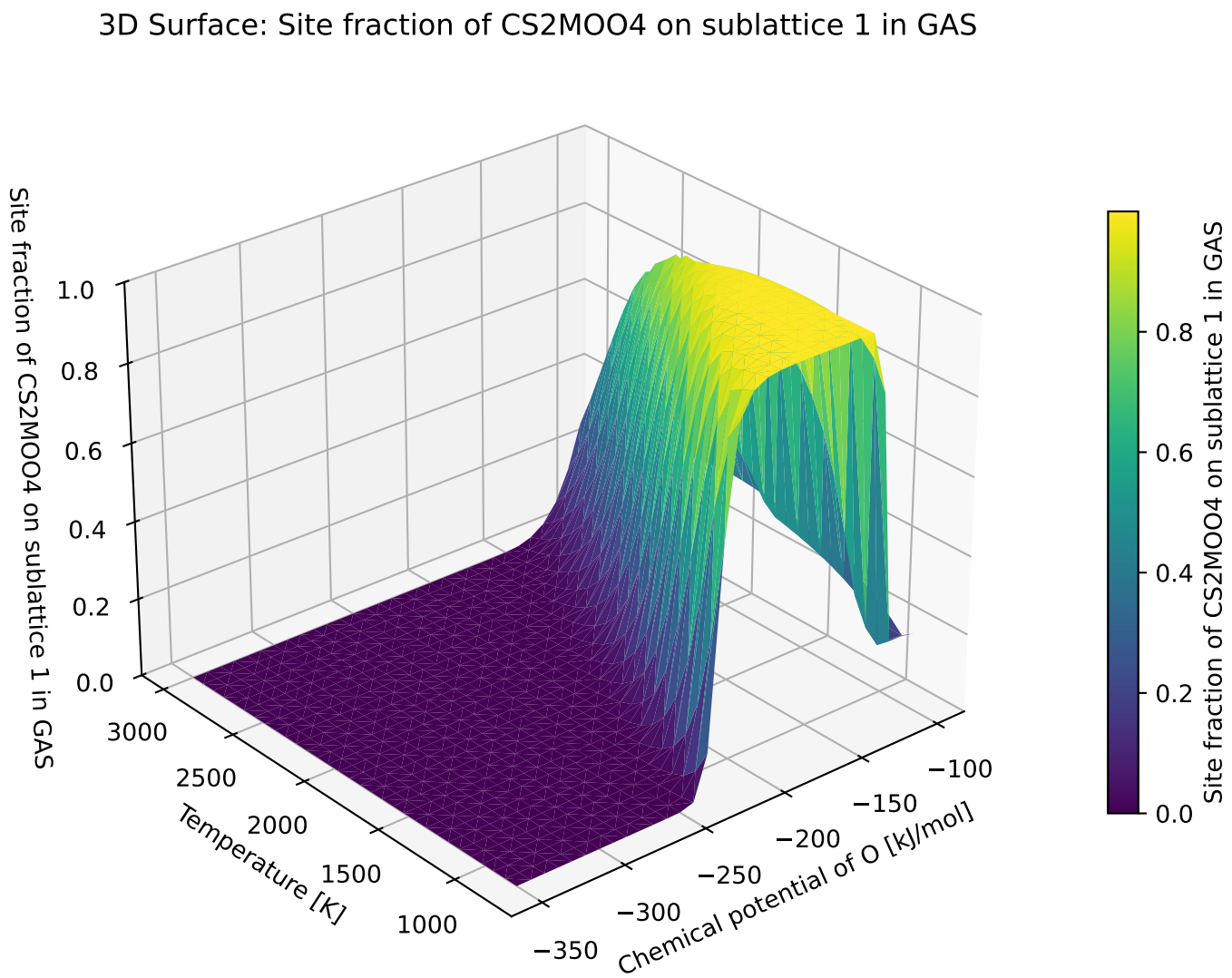

Contour: Mole fraction of O in GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of O in GAS
0.72
0.64
0.56
0.48
0.40
0.32
0.24
0.16
0.08
0.00

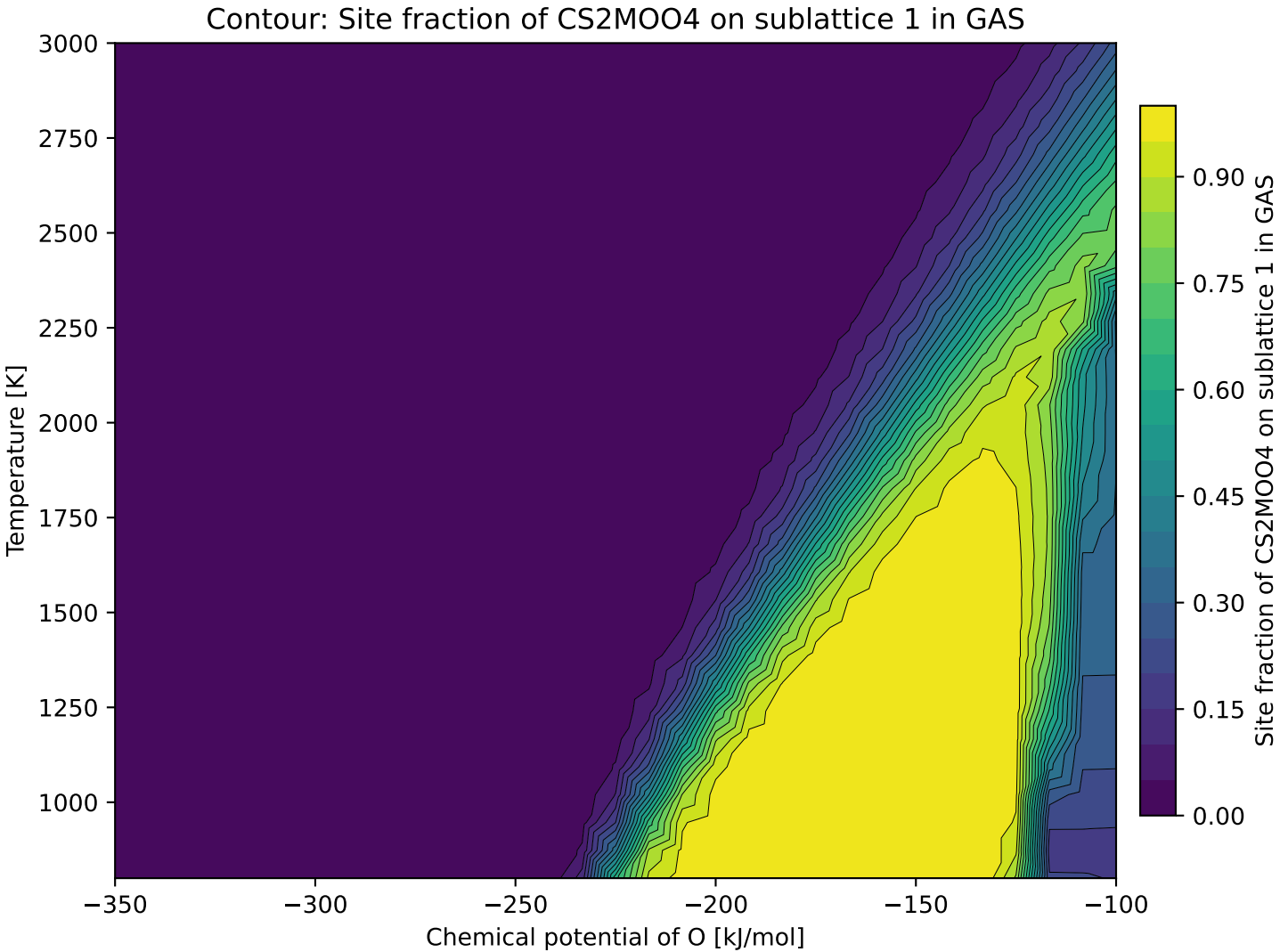

3D Surface: Mole fraction of Cs in GAS
Mole fraction of Cs in GAS
1.0
0.8
0.6
0.4
0.2
0.0
Temperature [K]
3000
2500
2000
1500
1000
Chemical potential of O [kJ/mol]
−350
−300
−250
−200
−150
−100
Mole fraction of Cs in GAS
1.0
0.8
0.6
0.4
0.2

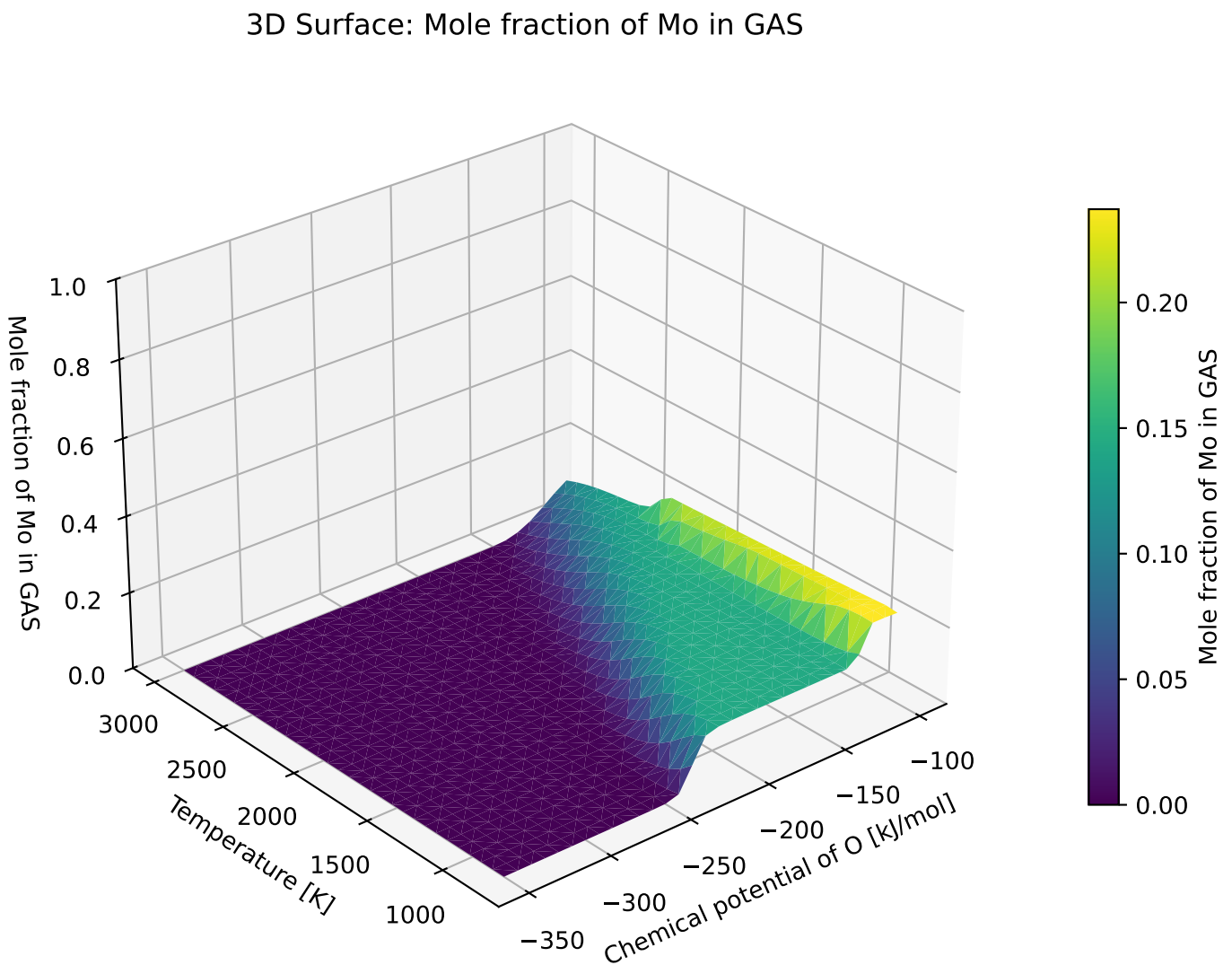

Contour: Mole fraction of Cs in GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of Cs in GAS
0.90
0.75
0.60
0.45
0.30
0.15
0.00


3D Surface: Site fraction of MOO3 on sublattice 1 in GAS
Site fraction of MOO3 on sublattice 1 in GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Site fraction of MOO3 on sublattice 1 in GAS

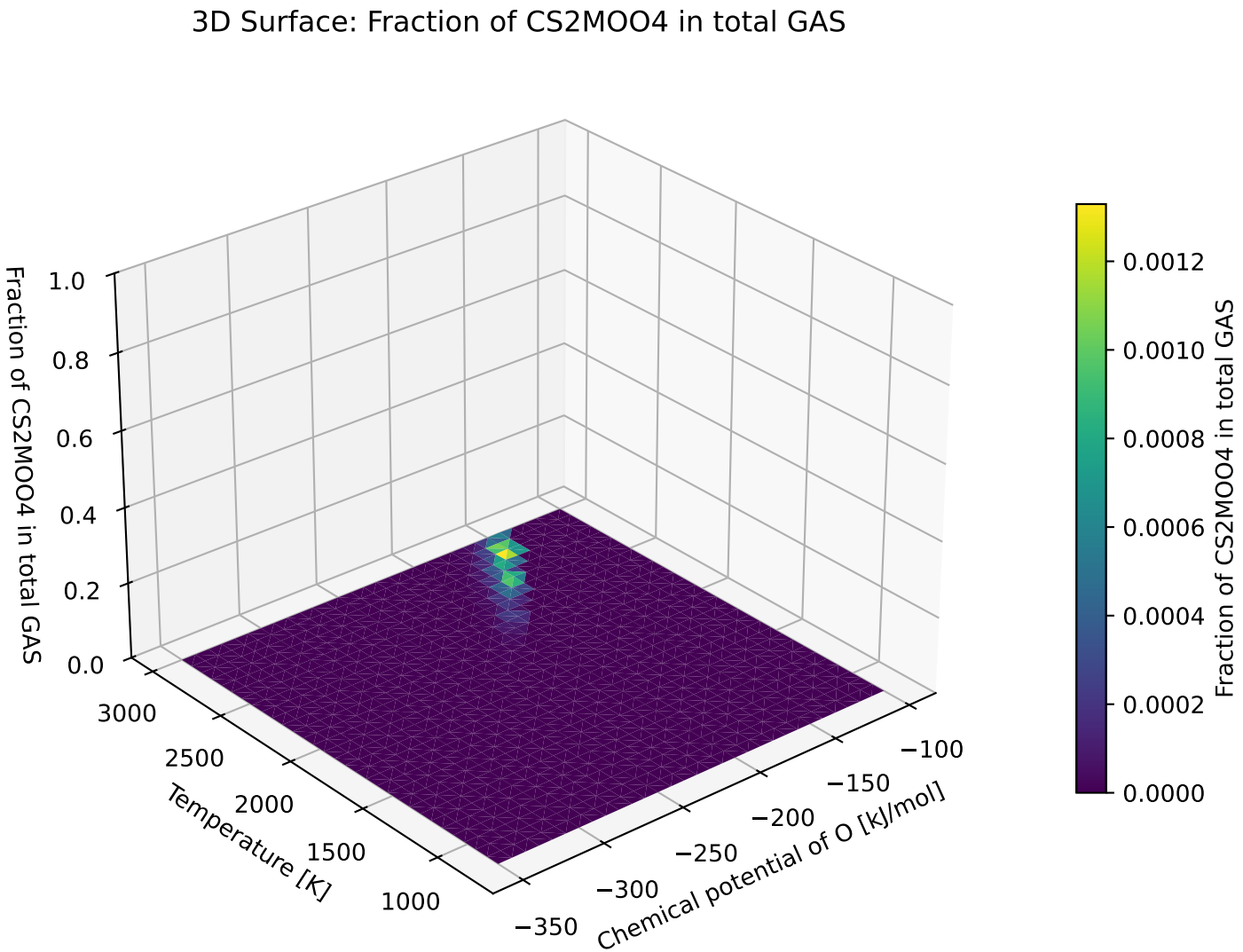

Contour: Site fraction of MOO3 on sublattice 1 in GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Site fraction of MOO3 on sublattice 1 in GAS
0.24
0.21
0.18
0.15
0.12
0.09
0.06
0.03
0.00

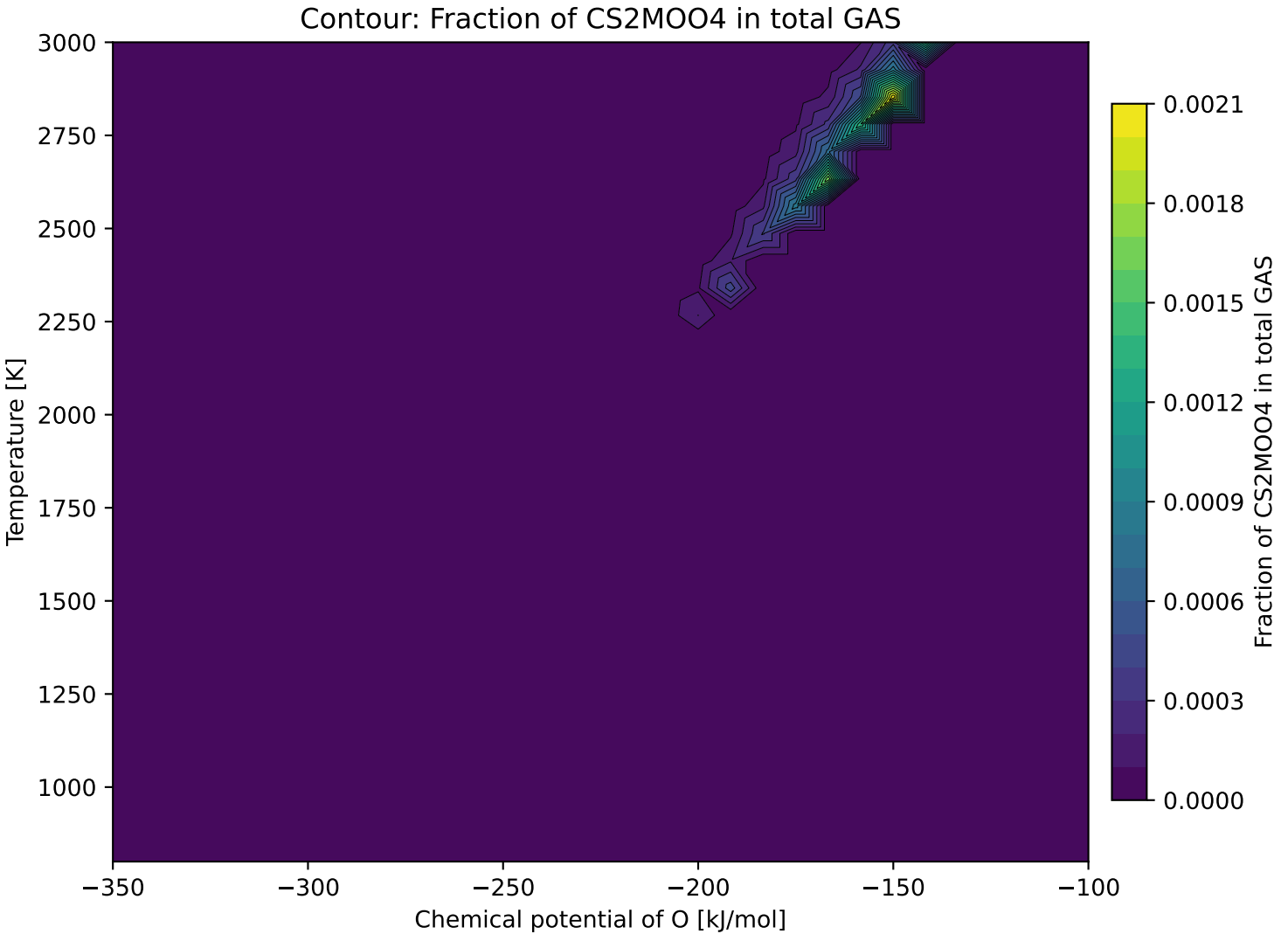

3D Surface: Site fraction of CS2MOO4 on sublattice 1 in GAS
Site fraction of CS2MOO4 on sublattice 1 in GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Site fraction of CS2MOO4 on sublattice 1 in GAS
0.8
0.6
0.4
0.2
0.0

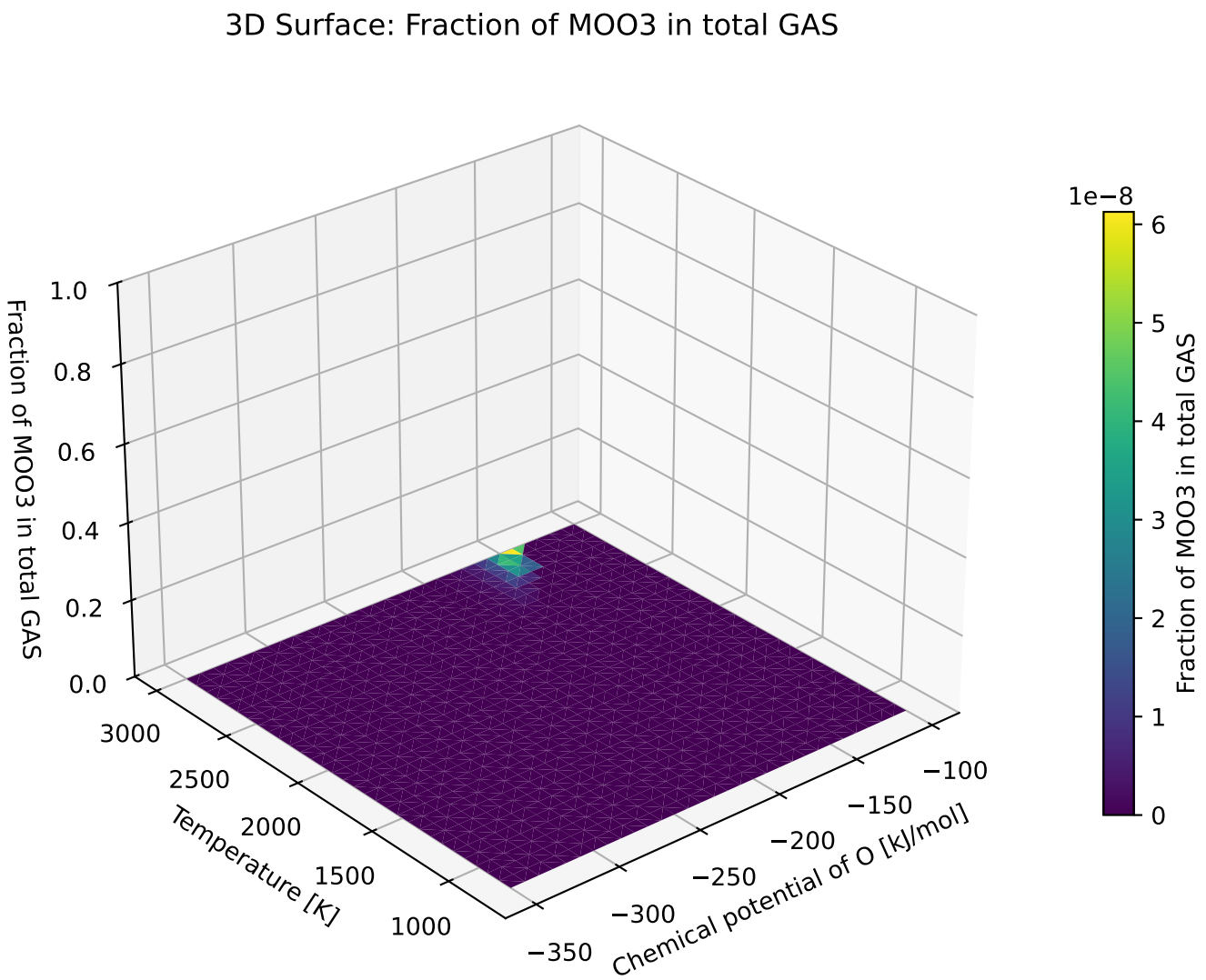

Contour: Site fraction of CS2MOO4 on sublattice 1 in GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Site fraction of CS2MOO4 on sublattice 1 in GAS
0.90
0.75
0.60
0.45
0.30
0.15
0.00

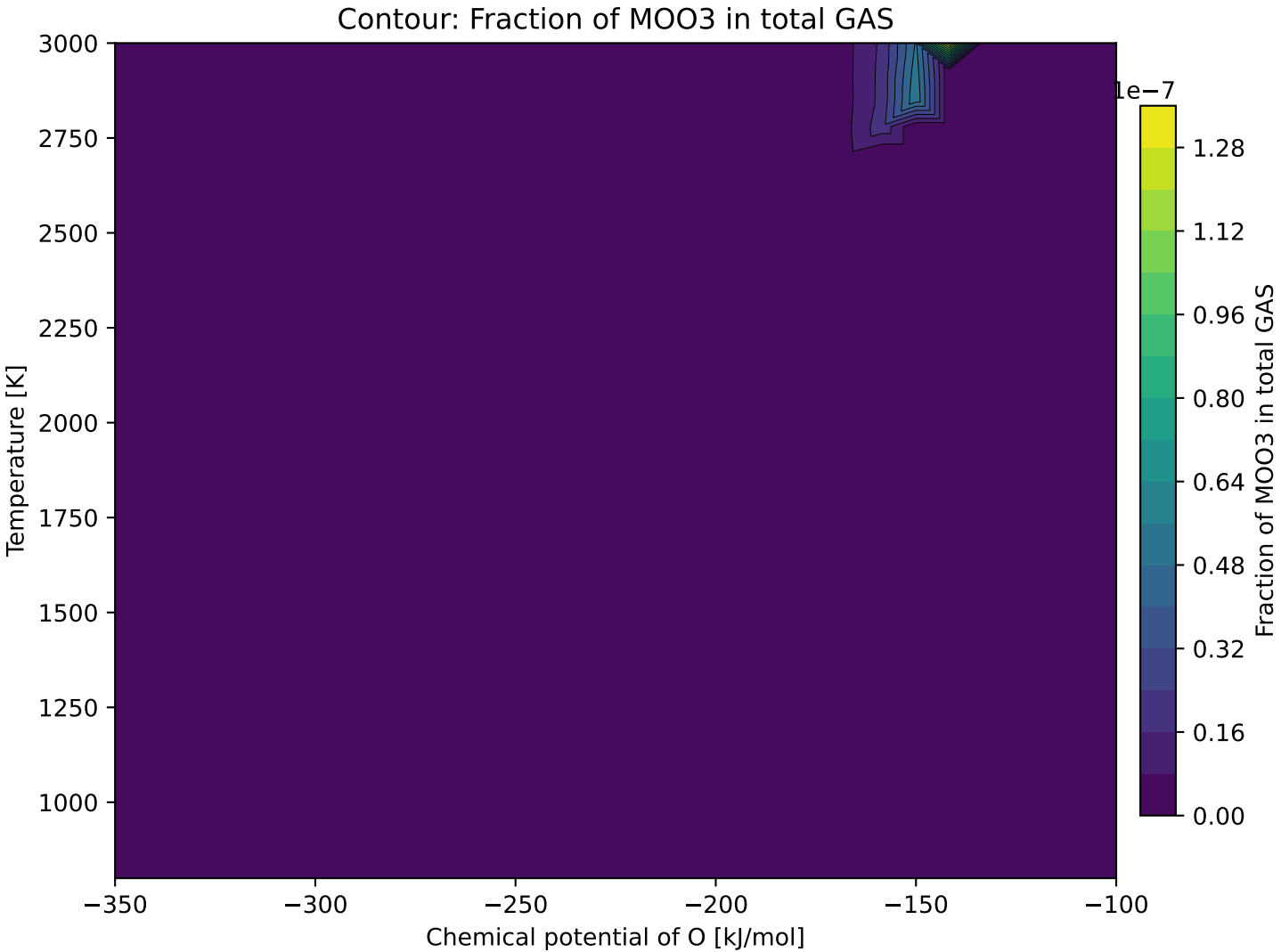

3D Surface: Mole fraction of Mo in GAS
Mole fraction of Mo in GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.20
0.15
0.10
0.05
0.00
Mole fraction of Mo in GAS

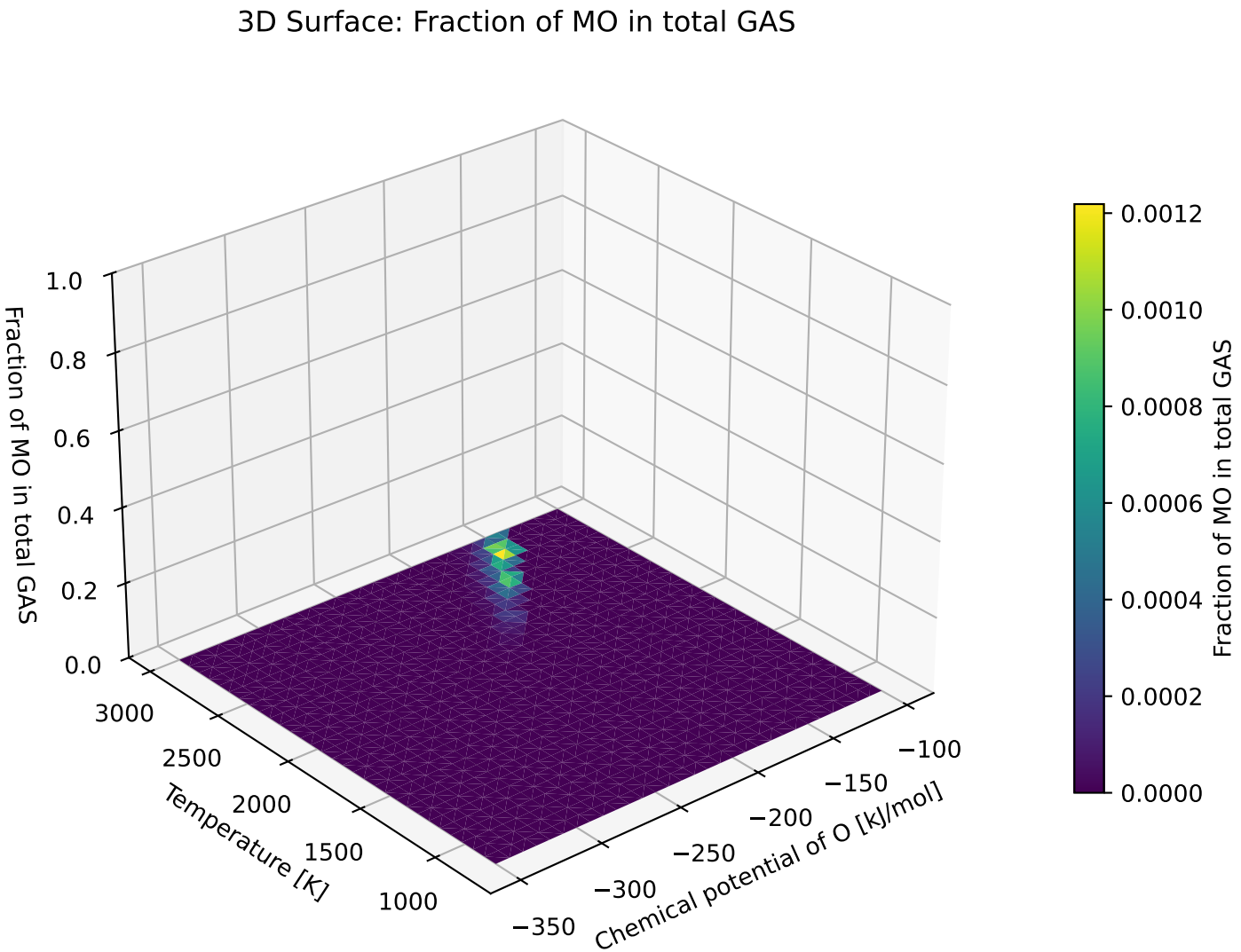
Contour: Mole fraction of Mo in GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Mole fraction of Mo in GAS
0.24
0.21
0.18
0.15
0.12
0.09
0.06
0.03
0.00

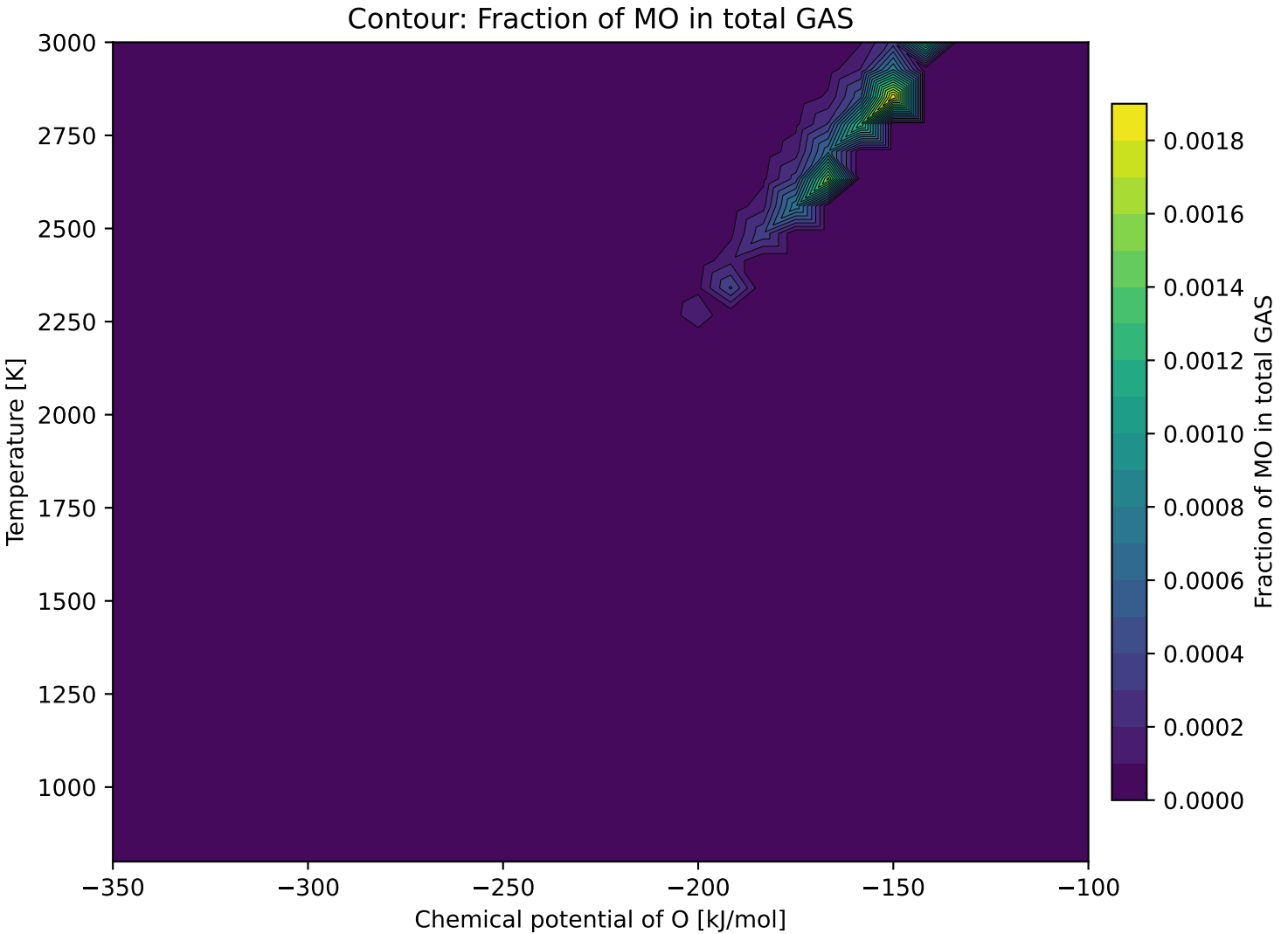

3D Surface: Fraction of CS2MOO4 in total GAS
Fraction of CS2MOO4 in total GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
Fraction of CS2MOO4 in total GAS


Contour: Fraction of CS2MOO4 in total GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Fraction of CS2MOO4 in total GAS
0.0021
0.0018
0.0015
0.0012
0.0009
0.0006
0.0003
0.0000


3D Surface: Fraction of MOO3 in total GAS
Fraction of MOO3 in total GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
1e−8
6
5
4
3
2
1
0
Fraction of MOO3 in total GAS


Contour: Fraction of MOO3 in total GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
1e−7
1.28
1.12
0.96
0.80
0.64
0.48
0.32
0.16
0.00
Fraction of MOO3 in total GAS


3D Surface: Fraction of MO in total GAS
Fraction of MO in total GAS
1.0
0.8
0.6
0.4
0.2
0.0
3000
2500
2000
1500
1000
Temperature [K]
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
Fraction of MO in total GAS


Contour: Fraction of MO in total GAS
Temperature [K]
3000
2750
2500
2250
2000
1750
1500
1250
1000
−350
−300
−250
−200
−150
−100
Chemical potential of O [kJ/mol]
Fraction of MO in total GAS
0.0018
0.0016
0.0014
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000